

Homophily in Congress on Twitter

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Introduction

We examine the Twitter Interaction Network for the 117th U.S. Congress by SNAP. The dataset contains 475 congressional Twitter accounts connected by 13,289 directed, weighted edges representing retweets, quote tweets, replies, and mentions. Edge weights capture the normalized influence between accounts as transmission probabilities. As the dataset contains no node attributes beyond usernames, we add party affiliation, chamber, age, and ideological scores from [Voteview](#), [github/unitedstates](#), and [USCD](#). We assess homophily across these characteristics, with particular interest in whether chamber and age moderate partisan homophily.

We assess homophily across these characteristics. It has been shown thoroughly that party homophily exists, but we are most interested in whether tertiary factors such as chamber and age moderate this effect, e.g. do members of the senate exhibit more homophily than those in the house, and, do older members exhibit more or less homophily than their younger colleagues?

Theory

Homophily is the principle that contact between similar people occurs at a higher rate than among dissimilar people. It can be further distinguished in two types: *status homophily* as the similarity of persons in their formal, informal or ascribed status and *value homophily* based on shared beliefs, values and attitudes of persons (McPherson et al. 2001). We take the party affiliation, the chamber and the age as status homophily and expect that the number of ties between members of congress is shaped by their status, e.g. congress members mostly interact with members of their own party. Age is expected to further shape the number of connections, since people tend to form ties with those of similar life stage (McPherson et al. 2001), older legislators may have formed cross-party

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relationships before the current era of polarization, but also older members may tend to be less active on twitter. Also a legislators chamber structures the opportunity for interaction, as Senate and House members operate in largely separate institutional environments with different rules, schedules, and legislative responsibilities, making cross-chamber ties less likely to form regardless of party. Value homophily is captured as left-right ideological position and stance on social and racial issues. There is a considerable tendency for adults to associate with those of their own political orientations (McPherson et al. 2001). Therefore, we expect to find more interactions between members with similar positioning in the ideological dimensions.

Assortativity is the network-level measure that formally captures homophily. Newman (2003) defines the assortativity coefficient as a Pearson correlation computed over all edges, ranging from -1 to 1 , where 1 indicates that all edges are within-group. In the context of the congressional Twitter network, assortativity provides a useful way to evaluate whether interaction patterns reflect political and demographic divisions. We therefore use assortativity to assess whether members cluster by party affiliation, chamber, age, and ideological position, and to compare the relative strength of these different forms of homophily within the network.

Data Pre-Processing

i Using `"',"` as decimal and `"'."` as grouping mark. Use ``read_delim()`` for more control.

```
# investigate NA values
vdf <- igraph::as_data_frame(g, what = "vertices")
na_counts <- colSums(is.na(vdf))

data.frame(
  `NA count` = na_counts,
  `NA %`      = round(na_counts / nrow(vdf) * 100, 1),
  check.names = FALSE
) |>
  arrange(desc(`NA count`))
```

	NA count	NA %
district	92	19.4
chamber	47	9.9
party	47	9.9
state	47	9.9
bioguide_id	47	9.9
icpsr	47	9.9
nominate_dim1	47	9.9
nominate_dim2	47	9.9
born	47	9.9

age24	47	9.9
full_name	5	1.1
name	0	0.0
username	0	0.0

Exploratory Data Analysis

Overall, the network is moderately sized and active, but not densely connected. Members form many retweet ties in absolute terms, yet these ties represent only a small share of all possible member-to-member connections ($\approx 5\%$). The typical congress member has around 22–24 connections, while a small number of members have far higher connectivity. This suggests a selective and uneven retweet structure, with activity concentrated around more central or more active members rather than spread uniformly across the network. This is to be expected given in politics there are often central figures that may receive more attention, such as presidents or speakers.

```
# Core descriptives: graph size

n_vertices <- vcount(g)
n_edges    <- ecount(g)
density    <- edge_density(g)

size_summary <- tibble::tibble(
  Measure = c(
    "Number of vertices",
    "Number of edges",
    "Edge density"
  ),
  Value = c(
    n_vertices,
    n_edges,
    density
  )
) %>%
  dplyr::mutate(Value = round(Value, 3))

cat(
  knitr::kable(
    size_summary,
    format = "pipe",
    col.names = c("Measure", "Value")
  ),
  sep = "\n"
)
```

Measure	Value
Number of vertices	475.000
Number of edges	13289.000
Edge density	0.059

```
# Degree and weighted degree / strength

deg_in <- degree(g, mode = "in")
deg_out <- degree(g, mode = "out")

str_in <- strength(g, mode = "in")
str_out <- strength(g, mode = "out")

degree_summary <- tibble::tibble(
  Measure = c(
    "In-degree",
    "Out-degree",
    "Weighted in-strength",
    "Weighted out-strength"
  ),
  Min = c(
    min(deg_in, na.rm = TRUE),
    min(deg_out, na.rm = TRUE),
    min(str_in, na.rm = TRUE),
    min(str_out, na.rm = TRUE)
  ),
  Q1 = c(
    quantile(deg_in, 0.25, na.rm = TRUE),
    quantile(deg_out, 0.25, na.rm = TRUE),
    quantile(str_in, 0.25, na.rm = TRUE),
    quantile(str_out, 0.25, na.rm = TRUE)
  ),
  Median = c(
    median(deg_in, na.rm = TRUE),
    median(deg_out, na.rm = TRUE),
    median(str_in, na.rm = TRUE),
    median(str_out, na.rm = TRUE)
  ),
  Mean = c(
    mean(deg_in, na.rm = TRUE),
    mean(deg_out, na.rm = TRUE),
    mean(str_in, na.rm = TRUE),
    mean(str_out, na.rm = TRUE)
  ),
  Q3 = c(
    quantile(deg_in, 0.75, na.rm = TRUE),
```

```

    quantile(deg_out, 0.75, na.rm = TRUE),
    quantile(str_in, 0.75, na.rm = TRUE),
    quantile(str_out, 0.75, na.rm = TRUE)
  ),
  Max = c(
    max(deg_in, na.rm = TRUE),
    max(deg_out, na.rm = TRUE),
    max(str_in, na.rm = TRUE),
    max(str_out, na.rm = TRUE)
  )
) %>%
  dplyr::mutate(
    dplyr::across(where(is.numeric), ~ round(.x, 3))
  )

cat(
  knitr::kable(
    degree_summary,
    format = "pipe"
  ),
  sep = "\n"
)

```

Measure	Min	Q1	Median	Mean	Q3	Max
In-degree	0.000	13.000	22.000	27.977	37.000	127.000
Out-degree	1.000	17.000	24.000	27.977	35.000	210.000
Weighted in-strength	0.000	0.058	0.117	0.163	0.204	1.648
Weighted out-strength	0.002	0.080	0.134	0.163	0.196	0.944

```

# Degree distributions

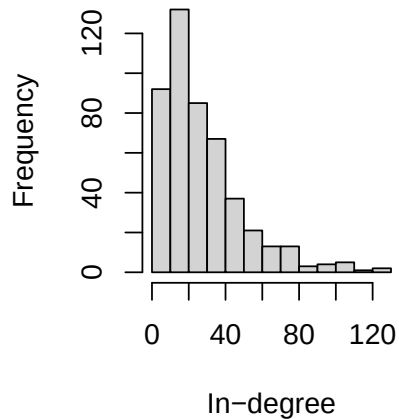
par(mfrow = c(1, 2))

hist(
  deg_in,
  main = "In-degree distribution",
  xlab = "In-degree"
)

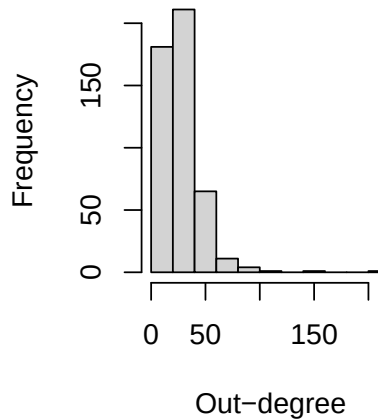
hist(
  deg_out,
  main = "Out-degree distribution",
  xlab = "Out-degree"
)

```

In-degree distribution



Out-degree distribution



```
par(mfrow = c(1, 1))

# Structural network measures

recip <- reciprocity(g)

trans_global <- transitivity(g, type = "global")
trans_avg <- transitivity(g, type = "average")

assort_degree <- assortativity_degree(g, directed = TRUE)

structure_summary <- tibble::tibble(
  Measure = c(
    "Reciprocity",
    "Global transitivity",
    "Average local transitivity",
    "Degree assortativity"
  ),
  Value = c(
    recip,
    trans_global,
    trans_avg,
    assort_degree
  )
) %>%
  dplyr::mutate(Value = round(Value, 3))

cat(
  knitr::kable(
```

```

    structure_summary,
    format = "pipe",
    col.names = c("Measure", "Value")
  ),
  sep = "\n"
)

```

Measure	Value
Reciprocity	0.462
Global transitivity	0.270
Average local transitivity	0.301
Degree assortativity	-0.096

The network shows a moderate level of reciprocity: about 46% of directed ties are reciprocated. This suggests that interactions are not purely one-sided, but often occur between pairs of members who engage with each other, possibly congress members working on the same topics or from similar geographic regions. Transitivity is also moderate, with global and average local clustering around 0.27–0.30, indicating that interaction patterns are somewhat clustered within groups of mutually connected members. This is to be expected given we are not accounting for chamber affiliation and or party affiliation at the moment, so clustering is to be expected. The slightly negative degree assortativity (-0.096) suggests that members with many outgoing interactions are not especially likely to direct those interactions toward members with many incoming interactions. In other words, highly active congress members are not disproportionately interacting with highly visible or frequently targeted congress members.

```

# Homophily by party
el <- igraph::as_data_frame(g, what = "edges")
el$from_party <- V(g)$party[match(el$from, V(g)$name)]
el$to_party <- V(g)$party[match(el$to, V(g)$name)]
mix <- xtabs(weight ~ from_party + to_party, data = el)
mix_prop <- mix / rowSums(mix)
cat("Overall\n", knitr::kable(round(mix_prop, 3), format =
  ↪ "pipe"), sep = "\n")

```

Overall

	Democrat	Independent	Republican
Democrat	0.874	0.001	0.125
Independent	0.985	0.000	0.015
Republican	0.123	0.000	0.877

We can see very strong party homophily: Democratic members mostly interact with Democrats (0.874), and Republican members mostly interact with

Republicans (0.877). Cross-party interaction is much lower, around 0.12 in both directions. The “Independent” category is too small to interpret. As we can see below, there does not seem to be much difference in these homophily structures between the house and senate.

```
# nominal assortativity
g_clean <- subgraph(g, which(!is.na(V(g)$party)))
party_int <- as.integer(as.factor(V(g_clean)$party))
assortativity_nominal(g_clean, types = party_int, directed =
  ↪ TRUE)
```

```
[1] 0.7150204
```

```
# Homophily by party by chamber
el$from_chamber <- V(g)$chamber[match(el$from, V(g)$name)]

for (ch in c("House", "Senate")) {
  el_ch <- el[!is.na(el$from_chamber) & el$from_chamber == ch, ]
  mix <- xtabs(weight ~ from_party + to_party, data = el_ch)
  mix_prop <- mix / rowSums(mix)
  cat(ch, "\n", knitr::kable(round(mix_prop, 3), format =
    ↪ "pipe"), sep = "\n", "\n")
}
```

House

	Democrat	Independent	Republican
Democrat	0.879	0.001	0.120
Independent	0.985	0.000	0.015
Republican	0.116	0.000	0.884

Senate

	Democrat	Independent	Republican
Democrat	0.826	0.003	0.170
Republican	0.201	0.000	0.799

Left-right ideology assortativity is similarly strong as that of party membership (0.739), meaning members tend to interact with ideologically similar members. Assortativity on the social/racial dimension is weaker (0.225), and age assortativity is essentially negligible (0.101) which means that congress members rarely interact more often with members of their age.


```

g_clean <- induced_subgraph(g, which(!is.na(V(g)$nominate_dim1)))
  ↪ # ensure only non-na values are assessed

# assortativity
leftrightass <- assortativity(g_clean, V(g_clean)$nominate_dim1,
  ↪ directed = TRUE)
socialracialass <- assortativity(g_clean,
  ↪ V(g_clean)$nominate_dim2, directed = TRUE)
ageass <- assortativity(g_clean, V(g_clean)$age24, directed =
  ↪ TRUE)

assort_table <- data.frame(dimension = c("Left-right",
  ↪ "Social/racial issues", "Age"), assortativity =
  ↪ c(leftrightass, socialracialass, ageass))

cat(
  knitr::kable(
    assort_table,
    digits = 3,
    col.names = c("Dimension", "Assortativity"),
    format = "pipe"
  ),
  sep = "\n"
)

```

Dimension	Assortativity
Left-right	0.739
Social/racial issues	0.225
Age	0.101

```

# Homophily by age
el$same_party <- !is.na(el$from_party) &
  !is.na(el$to_party) &
  el$from_party == el$to_party

# filter NAs
el_known <- el %>% filter(!is.na(from_party), !is.na(to_party))
# exclude Independents
el_known <- el_known[el_known$from_party != "Independent" &
  ↪ el_known$to_party != "Independent", ]

# per-node homophily score
homophily_score <- el_known %>%
  group_by(from) %>%
  summarise(total_w = sum(weight), ingroup_w =
    ↪ sum(weight[same_party]), h_score = ingroup_w / total_w)

```

```

homophily_score$age <- V(g)$age24[match(homophily_score$from,
  ↪ V(g)$name)]
homophily_score$party <- V(g)$party[match(homophily_score$from,
  ↪ V(g)$name)]

# correlation
cat("Correlation between age and
  ↪ homophily:\n", cor(homophily_score$h_score,
  ↪ homophily_score$age, use = "complete.obs"), "\n\n")

```

Correlation between age and homophily:
-0.03806106

```

# by party --> democrats weakly positive (+0.04), republicans
  ↪ kind of substantially negative (-0.128)
homophily_score %>%
  group_by(party) %>%
  summarise(`Correlation Age <-> Party Homophily` = cor(h_score,
  ↪ age, use = "complete.obs")) %>%
  mutate(`Correlation Age <-> Party Homophily` =
  ↪ round(`Correlation Age <-> Party Homophily`, 3)) %>%
  { cat(knitr::kable(., format = "pipe"), sep = "\n") }

```

party	Correlation Age <-> Party Homophily
:-----	-----:
Democrat	0.058
Republican	-0.138

Here, we are looking at whether the party homophily of individuals is higher or lower depending on their age. The correlation is very small and slightly negative (-0.038) overall, suggesting that older and younger members do not differ much in their tendency to interact within party. By party, the relationship is weakly positive for Democrats (0.058) and modestly negative for Republicans (-0.138), indicating that older Republican members are somewhat less party-homophilous in their tweet interactions.

Visualization

```

g_party <- subgraph(g, which(!is.na(V(g)$party)))
no_of_nodes <- vcount(g_party)

set.seed(7341)

# stress layout with inverted edge weights

```

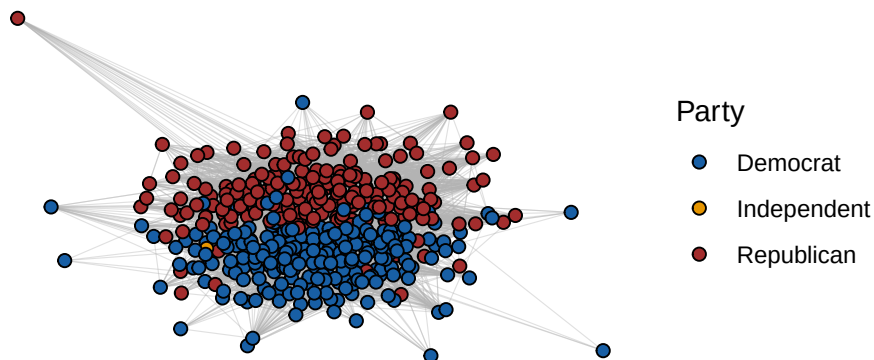
```

ggraph(g_party, layout = "stress", weights = 1 /
  ↪ E(g_party)$weight) +
  geom_edge_link0(edge_colour = "grey70", edge_alpha = 0.4,
    ↪ edge_width = 0.2) +
  geom_node_point(aes(fill = party), shape = 21, size = 2) +
  scale_fill_manual(
    values = c("Democrat" = "#185FA5", "Republican" = "#A32D2D",
      "Independent" = "#E69500"),
    name = "Party"
  ) +
  theme_graph(base_family = "") +
  labs(title = "Stress Layout Overview",
    subtitle = paste0("Nodes with known party only (n = ",
      ↪ no_of_nodes, ")"))

```

Stress Layout Overview

Nodes with known party only (n = 428)



```

set.seed(7341)

# backbone layout needs an undirected graph so we collapse edges
↪ using the mean of in and out
g_party_un <- as_undirected(g_party, mode = "collapse",
  ↪ edge.attr.comb = "mean")

ggraph(g_party_un, layout = "backbone") +
  geom_edge_link0(edge_colour = "grey70", edge_alpha = 0.4,
    ↪ edge_width = 0.2) +
  geom_node_point(aes(fill = party), shape = 21, size = 2) +
  scale_fill_manual(
    values = c("Democrat" = "#185FA5", "Republican" = "#A32D2D",
      "Independent" = "#E69500"),

```

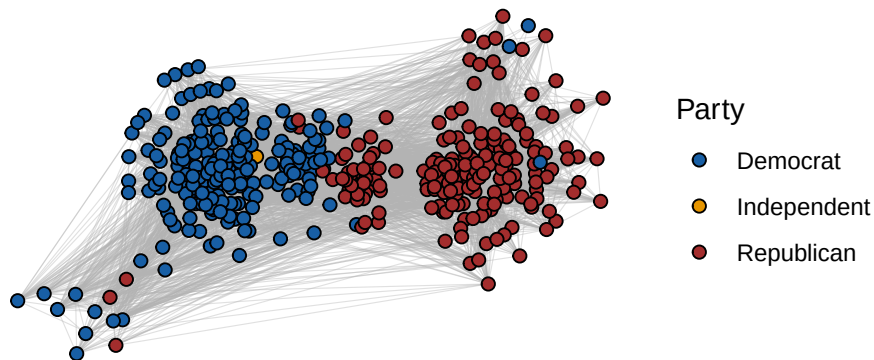
```

    name = "Party"
  ) +
  theme_graph(base_family = "") +
  labs(title = "Backbone Layout Overview",
       subtitle = paste0("Nodes with known party only (n = ",
                          no_of_nodes, ")"))

```

Backbone Layout Overview

Nodes with known party only (n = 428)



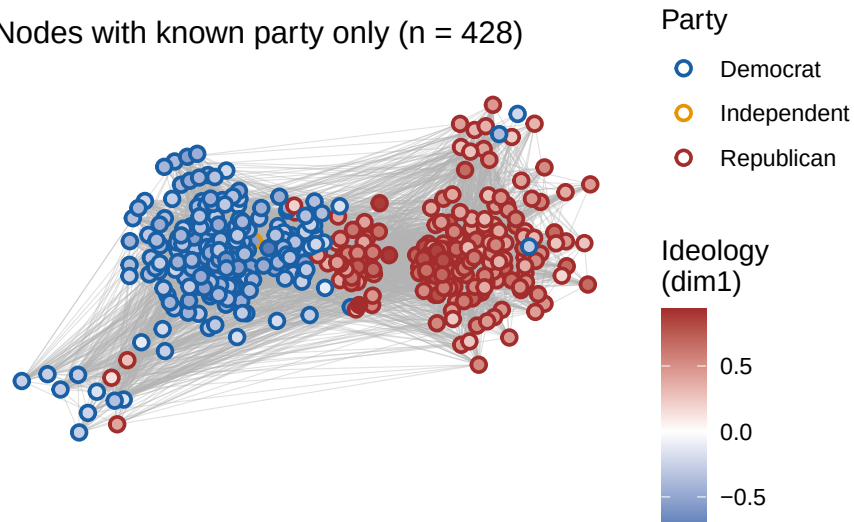
```

# fill is a gradient by ideology dim1, party is the color contour
ggraph(g_party_un, layout = "backbone") +
  geom_edge_link0(edge_colour = "grey70", edge_alpha = 0.4,
    edge_width = 0.2) +
  geom_node_point(aes(fill = nominate_dim1, colour = party),
    shape = 21, size = 2, stroke = 1) +
  scale_fill_gradient2(
    low = "#185FA5",
    mid = "white",
    high = "#A32D2D",
    midpoint = 0,
    name = "Ideology\n(dim1)"
  ) +
  scale_colour_manual(
    values = c("Democrat" = "#185FA5", "Republican" = "#A32D2D",
              "Independent" = "#E69500"),
    name = "Party"
  ) +
  theme_graph(base_family = "") +
  labs(title = "Backbone Layout Overview with Ideology",
       subtitle = paste0("Nodes with known party only (n = ",
                          no_of_nodes, ")"))

```

Backbone Layout Overview with Ideology

Nodes with known party only (n = 428)



Both layouts confirm strong party homophily with few cross-party exceptions. The single independent node (Bernie Sanders) interacts largely with Democrats as expected. Backbone also shows a divide both in the Democratic and Republican party not visible in Stress. The latter divide acts as a bridge between the parties. The inclusion of ideology (dim1) also shows according clustering, with higher values closer together, particularly for the Republicans.

```
max_node <- which.max(degree(g, mode = "out"))
cat("Username :", V(g)$username[max_node], "\n")
```

Username : SpeakerPelosi

```
cat("Full name:", V(g)$full_name[max_node], "\n")
```

Full name: Nancy Pelosi

```
cat("Party :", V(g)$party[max_node], "\n")
```

Party : Democrat

```
cat("Age :", V(g)$age24[max_node], "\n")
```

Age : 84

```
cat("Out-degree:", degree(g, mode = "out")[max_node], "\n")
```

Out-degree: 210

```
cat("Out-strength:", round(strength(g, mode = "out")[max_node],
↪ 4), "\n")
```

Out-strength: 0.9436

```
focal_name <- V(g)$name[V(g)$username == "SpeakerPelosi"]
ego_g <- make_ego_graph(g, order = 1,
  nodes = which(V(g)$name ==
    ↪ focal_name),
  mode = "all")[[1]]
focal_idx_ego <- which(V(ego_g)$name == focal_name)

# Top 10 out-edges
eligible_eids <- as.integer(incident(ego_g, focal_idx_ego, mode =
  ↪ "out"))
top10_out_idx <-
  ↪ eligible_eids[order(E(ego_g)$weight[eligible_eids],
  ↪ decreasing = TRUE)][1:10]
ego_out <- subgraph_from_edges(ego_g, eids = top10_out_idx,
  ↪ delete.vertices = TRUE)
ego_out <- induced_subgraph(ego_out,
  ↪ which(!is.na(V(ego_out)$party)))
focal_idx2 <- which(V(ego_out)$name == focal_name)

ggraph(ego_out, layout = "focus", focus = focal_idx2) +
  draw_circle(col = "grey80", use = "focus", max.circle = 1) +
  geom_edge_link0(aes(edge_width = weight), edge_colour =
    ↪ "grey25", edge_alpha = 0.5) +
  scale_edge_width_continuous(name = "Transmission\nprobability",
    ↪ range = c(0.3, 2.5)) +
  geom_node_point(aes(fill = party), shape = 21, size = 3) +
  scale_fill_manual(
    values = c("Democrat" = "#185FA5", "Republican" =
      ↪ "#A32D2D",
      "Independent" = "#534AB7"),
    na.value = "grey80",
    name = "Party"
  ) +
  geom_node_text(aes(label = full_name), size = 2.5, vjust = -1)
  ↪ +
  geom_node_text(
    aes(label = round(nominate_dim1, 2)),
    size = 2.2, vjust = 2, colour = "black"
  ) +
  theme_graph(base_family = "") +
  coord_cartesian(clip = "off") +
  labs(title = paste("Ego network:",
    ↪ V(ego_out)$full_name[focal_idx2]),
```

```

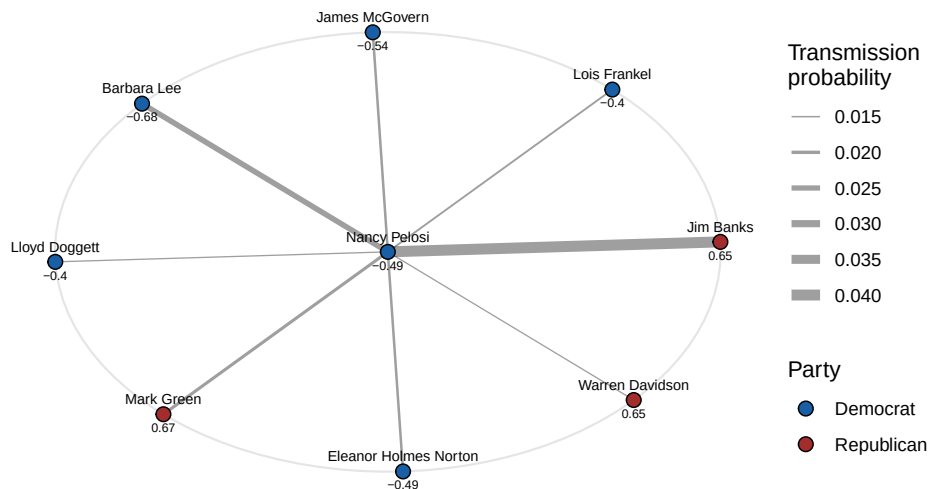
subtitle = "Top 10 out-edges by transmission probability.
  ↳ \nValues below nodes indicate DW-Nominate dimension
  ↳ 1.")

```

Warning: `aes\_()` was deprecated in ggplot2 3.0.0.
 i Please use tidy evaluation idioms with `aes()`
 i The deprecated feature was likely used in the graphlayouts package.
 Please report the issue at
<https://github.com/schochastics/graphlayouts/issues>.

Ego network: Nancy Pelosi

Top 10 out-edges by transmission probability.
 Values below nodes indicate DW-Nominate dimension 1.



We look at the top 10 edges of an example node, Nancy Pelosi, in an ego network. She has the highest out-degree of all nodes. Despite the majority of the nodes following party and ideology homophily, there are major exceptions, particularly with Jim Banks who has an ideological distance of 1.14 to Pelosi. This is likely a result of conflict interaction rather than collaboration. This illustrates that aggregate homophily patterns can coexist with meaningful individual-level violations, particularly among high-visibility leadership figures.

```

el <- igraph::as_data_frame(g, what = "edges")
el$from_party <- V(g)$party[match(el$from, V(g)$name)]
el$to_party <- V(g)$party[match(el$to, V(g)$name)]
el$from_dim1 <- V(g)$nominate_dim1[match(el$from, V(g)$name)]
el$to_dim1 <- V(g)$nominate_dim1[match(el$to, V(g)$name)]
el$ideo_distance <- abs(el$from_dim1 - el$to_dim1)

assort_by_bin <- el %>%

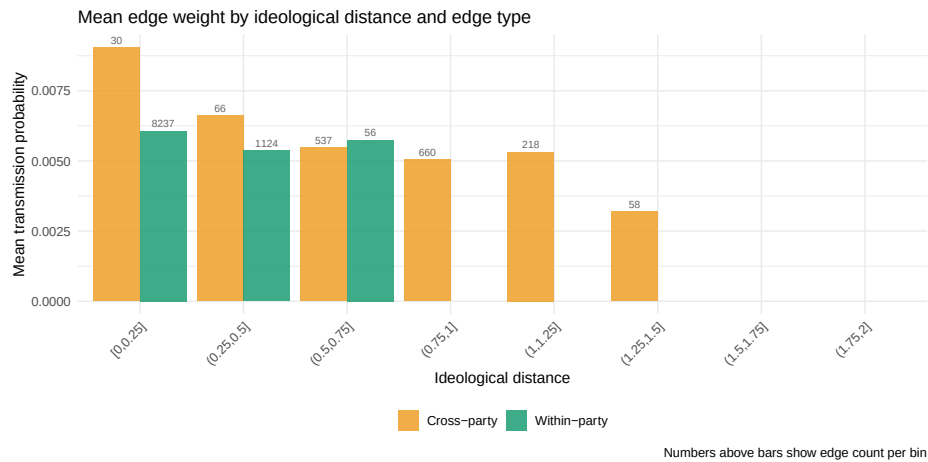
```

```

filter(!is.na(from_party), !is.na(to_party),
  ↳ !is.na(ideo_distance)) %>%
mutate(
  edge_type    = ifelse(from_party == to_party, "Within-party",
  ↳ "Cross-party"),
  distance_bin = cut(ideo_distance, breaks = seq(0, 2, by =
  ↳ 0.25),
                                include.lowest = TRUE)
) %>%
group_by(edge_type, distance_bin) %>%
summarise(n_edges = n(), mean_weight = mean(weight), .groups =
  ↳ "drop") %>%
filter(!is.na(distance_bin)) %>%
tidyr::complete(edge_type, distance_bin) # fills missing
↳ combinations with NA

ggplot(assort_by_bin, aes(x = distance_bin, y = mean_weight, fill
  ↳ = edge_type)) +
  geom_col(position = position_dodge(width = 0.9), alpha = 0.85,
  ↳ na.rm = TRUE) +
  geom_text(aes(label = n_edges), position = position_dodge(width
  ↳ = 0.9),
            vjust = -0.5, size = 2.8, colour = "grey40", na.rm =
  ↳ TRUE) +
  scale_fill_manual(values = c("Within-party" = "#1D9E75",
  ↳ "Cross-party" = "#EF9F27")) +
  labs(
    title    = "Mean edge weight by ideological distance and edge
    ↳ type",
    x        = "Ideological distance",
    y        = "Mean transmission probability",
    fill     = NULL,
    caption  = "Numbers above bars show edge count per bin"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x    = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )

```

```
# we look at why there are such high edge weights across parties
↳ for low ideological distance
# and find it is entirely Bernie Sanders so we remove
↳ independents from our analysis here
el %>%
  filter(!is.na(from_party), !is.na(to_party),
    ↳ !is.na(ideo_distance)) %>%
  mutate(edge_type = ifelse(from_party == to_party,
    ↳ "Within-party", "Cross-party"),
    distance_bin = cut(ideo_distance, breaks = seq(0, 2, by
    ↳ = 0.25),
    include.lowest = TRUE)) %>%
  filter(edge_type == "Cross-party", distance_bin == "[0,0.25]")
  ↳ %>%
  left_join(tibble(from = V(g)$name, from_name = V(g)$full_name),
    ↳ by = "from") %>%
  left_join(tibble(to = V(g)$name, to_name = V(g)$full_name),
    ↳ by = "to") %>%
  select(from_party, to_party, from_name, to_name, ideo_distance,
    ↳ weight) %>%
  arrange(ideo_distance)
```

	from_party	to_party	from_name	to_name
1	Independent	Democrat	Bernard Sanders	James McGovern
2	Independent	Democrat	Bernard Sanders	Jimmy Gomez
3	Independent	Democrat	Bernard Sanders	Mark Pocan
4	Independent	Democrat	Bernard Sanders	Edward Markey
5	Independent	Democrat	Bernard Sanders	Bonnie Watson Coleman
6	Independent	Democrat	Bernard Sanders	Yvette Clarke
7	Independent	Democrat	Bernard Sanders	Pramila Jayapal
8	Independent	Democrat	Bernard Sanders	Judy Chu

9	Independent	Democrat	Bernard Sanders	Andy Levin
10	Independent	Democrat	Bernard Sanders	Sean Casten
11	Independent	Democrat	Bernard Sanders	Ro Khanna
12	Independent	Democrat	Bernard Sanders	Jeff Merkley
13	Independent	Democrat	Bernard Sanders	Alex Padilla
14	Independent	Democrat	Bernard Sanders	Jesús "Chuy" García
15	Independent	Democrat	Bernard Sanders	Jared Huffman
16	Independent	Democrat	Bernard Sanders	Peter DeFazio
17	Independent	Democrat	Bernard Sanders	Barbara Lee
18	Independent	Democrat	Bernard Sanders	Jason Crow
19	Democrat	Independent	Jason Crow	Bernard Sanders
20	Independent	Democrat	Bernard Sanders	Patrick Leahy
21	Independent	Democrat	Bernard Sanders	Jamaal Bowman
22	Democrat	Independent	Jamaal Bowman	Bernard Sanders
23	Independent	Democrat	Bernard Sanders	Eric Swalwell
24	Independent	Democrat	Bernard Sanders	Elizabeth Warren
25	Democrat	Independent	Elizabeth Warren	Bernard Sanders
26	Independent	Democrat	Bernard Sanders	Brad Sherman
27	Independent	Democrat	Bernard Sanders	Sheldon Whitehouse
28	Democrat	Independent	Sheldon Whitehouse	Bernard Sanders
29	Republican	Democrat	Brian Fitzpatrick	Jared Golden
30	Independent	Democrat	Bernard Sanders	Ilhan Omar

	ideo_distance	weight
1	0.007	0.0047393365
2	0.015	0.0047393365
3	0.024	0.0094786730
4	0.028	0.0142180095
5	0.049	0.0142180095
6	0.062	0.0047393365
7	0.065	0.0094786730
8	0.066	0.0047393365
9	0.071	0.0142180095
10	0.083	0.0047393365
11	0.086	0.0236966825
12	0.091	0.0047393365
13	0.102	0.0142180095
14	0.107	0.0236966825
15	0.113	0.0047393365
16	0.118	0.0047393365
17	0.131	0.0094786730
18	0.181	0.0142180095
19	0.181	0.0031545741
20	0.185	0.0142180095
21	0.188	0.0236966825
22	0.188	0.0016778523
23	0.190	0.0047393365

```

24         0.197 0.0142180095
25         0.197 0.0019267823
26         0.200 0.0047393365
27         0.204 0.0094786730
28         0.204 0.0009718173
29         0.248 0.0029850746
30         0.249 0.0047393365

```

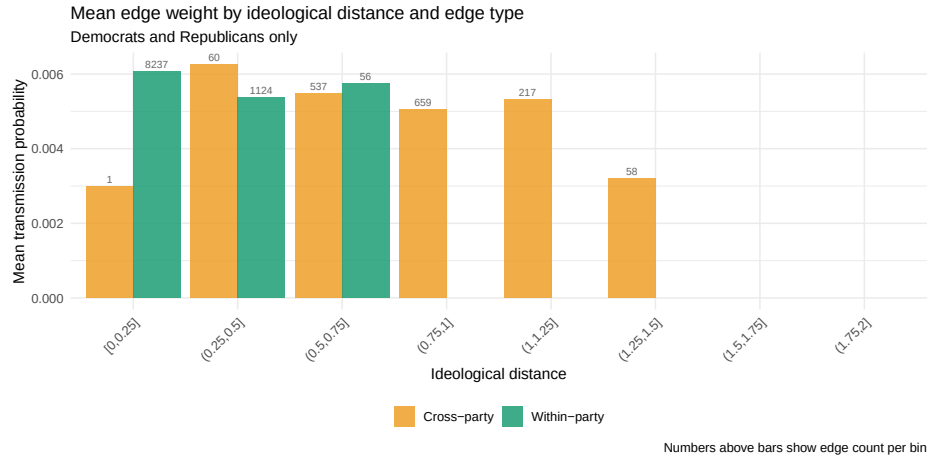
```

assort_by_bin <- el %>%
  filter(!is.na(from_party), !is.na(to_party),
    ↪ !is.na(ideo_distance),
    from_party != "Independent", to_party != "Independent")
↪ %>% # exclude here
mutate(
  edge_type = ifelse(from_party == to_party, "Within-party",
    ↪ "Cross-party"),
  distance_bin = cut(ideo_distance, breaks = seq(0, 2, by =
    ↪ 0.25),
    include.lowest = TRUE)
) %>%
group_by(edge_type, distance_bin) %>%
summarise(n_edges = n(), mean_weight = mean(weight), .groups =
  ↪ "drop") %>%
filter(!is.na(distance_bin)) %>%
tidyr::complete(edge_type, distance_bin)

ggplot(assort_by_bin, aes(x = distance_bin, y = mean_weight, fill
  ↪ = edge_type)) +
  geom_col(position = position_dodge(width = 0.9), alpha = 0.85,
    ↪ na.rm = TRUE) +
  geom_text(aes(label = n_edges), position = position_dodge(width
    ↪ = 0.9),
    vjust = -0.5, size = 2.8, colour = "grey40", na.rm =
    ↪ TRUE) +
  scale_fill_manual(values = c("Within-party" = "#1D9E75",
    ↪ "Cross-party" = "#EF9F27")) +
  labs(
    title = "Mean edge weight by ideological distance and edge
    ↪ type",
    subtitle = "Democrats and Republicans only",
    x = "Ideological distance",
    y = "Mean transmission probability",
    fill = NULL,
    caption = "Numbers above bars show edge count per bin"
  ) +
  theme_minimal(base_size = 12) +

```

```
theme(
  axis.text.x      = element_text(angle = 45, hjust = 1),
  legend.position  = "bottom"
)
```



Looking at mean edge weights by ideological distance, we find the expected pattern of high interaction within parties on low ideological distance. Interestingly, a similarly high interaction persists across parties even as ideological distance increases. Only at the highest distance bin $([1.25, 1.5])$ does it decrease. On an aggregate level, meaningful interaction across parties exists up to a threshold. Though, again, this is likely more confrontational than collaborative. We might expect interaction due to confrontation to increase again at high ideological differences, but we do not observe this. Transmission probability does not increase at high distances, suggesting that conflict-driven Twitter engagement, while present, does not dominate aggregate interaction patterns.

Conclusion

Results consistently confirm strong partisan homophily: members direct approximately 87% of interaction weight toward co-partisans, symmetrically across both parties and chambers. Ideological assortativity on the left-right dimension reinforces this pattern, while the social-racial dimension and age produce much weaker effects. Older Republican members show a modest tendency toward less party-homophilous interaction, though the overall age effect is negligible. Network visualizations confirm partisan clustering, with dense clusters of similar ideology. However, we also find exceptions, such as Pelosi's strong tie to Jim Banks. Such ties likely reflect conflict interaction rather than cross-party collaboration.

Key limitations include 47 unmatched nodes, the limitation to the 117th Congress,

and the inability to distinguish collaborative from adversarial interactions in the transmission probability metric. Future research could incorporate interaction type classification to separate confrontational from collaborative ties, integrate committee co-membership data as a structural covariate, or extend the analysis across multiple congresses to assess whether homophily has intensified alongside broader political polarization.

References

- McPherson, Miller, Lynn Smith-Lovin, and James M. Cook. 2001. “Birds of a Feather: Homophily in Social Networks.” *Annual Review of Sociology* 27 (1): 415–44. <https://doi.org/10.1146/annurev.soc.27.1.415>.
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